

14. Compound Interest

1) Simple Interest (I) = $\frac{P \cdot N \cdot R}{100}$

2) Compound Interest :-

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$\therefore \text{Compound Interest} = A - P$$

Here,

I = Interest

A = Amount

P = Principle

R = Rate of interest

N = Number of years

Practice set 14.1

1. Find the amount and the compound interest.

No.	Principle (₹)	Rate (p.c.p.a)	Duration (years)
1.	2000	5	2
2.	5000	8	3
3.	4000	7.5	2

1. $P = 2000$, $R = 5\%$, $N = 2 \text{ years}$

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$= 2000 \left[1 + \frac{5}{100} \right]^2$$

$$= 2000 \left[1 + \frac{1}{20} \right]^2$$

$$= 2000 \left[\frac{20+1}{20} \right]$$

$$= 2000 \left[\frac{21}{20} \right]^2$$

$$= 2000 \times \frac{441}{400}$$

$$A = 5 \times 441$$

2. $P = 5000$, $R = 8\%$, $N = 3 \text{ years}$

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$A = 5000 \left[1 + \frac{8}{100} \right]^3$$

$$A = 5000 \times \left[\frac{1}{1} + \frac{2}{25} \right]^3$$

$$A = 5000 \times \left[\frac{25+2}{25} \right]^3$$

$$A = 5000 \times \left[\frac{27}{25} \right]^3$$

$$A = \overset{8}{200} \times \frac{27}{25} \times \frac{27}{25} \times \frac{27}{25}$$

$$A = \frac{8 \times 27 \times 27 \times 27}{25} = \frac{216 \times 27 \times 27}{25}$$

$$A = \frac{157464}{25}$$

$$A = 6298.5$$

3. $P = 4000$, $R = 7.5$, $N = 2$

$$\Rightarrow A = P \left[1 + \frac{R}{100} \right]^N$$

$$A = 4000 \times \left[\frac{1}{1} + \frac{7.5}{100} \right]^2$$

$$A = 4000 \times \left[\frac{1}{1} + \frac{0.3}{4} \right]^2$$

$$A = 4000 \times \left[\frac{4 + 0.3}{4} \right]^2$$

$$A = 4000 \times \left[\frac{4.3}{4} \right]^2$$

$$A = \overset{250}{\cancel{4000}} \times 4.3 \times 4.3 = 250 \times 4.3 \times 4.3$$

$4 \times A_1$

$$A = 4622.5$$

2. Sameer has taken a loan of ₹ 12500 at a rate of 12% p.a. for 3 years. If the interest is compounded annually then how many rupees should he pay to clear his loan?

$$\Rightarrow P = 12500, R = 12\%, N = 3 \text{ years}$$

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$A = 12500 \left[\frac{1}{1} + \frac{12}{100} \right]^3$$

$\frac{12}{25}$

$$A = 12500 \left[\frac{1}{1} + \frac{3}{25} \right]^3$$

$$A = 12500 \left[\frac{25+3}{25} \right]^3$$

$$A = 12500 \left[\frac{28}{25} \right]^3$$

$$A = \frac{12500 \times 28 \times 28 \times 28}{25 \times 25 \times 25}$$

$$A = \frac{560 \times 28 \times 28}{25}$$

$$A = \frac{15680 \times 28}{25}$$

$$A = \frac{392000}{25} = 15680$$

$$A = 15680 \times 1.05 = 17561.6$$

17561.6 rupees should he pay to clear his loan.

3. To start a business Shalaka has taken a loan of ₹ 8000 at a rate of $10\frac{1}{2}$ p.c.p.a. After two years how much compound interest will she have to pay?

⇒

Given - $P = 8000$, $R = 10\frac{1}{2}\% = 10.5\%$,
 $N = 2$ years

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$A = 8000 \left[1 + \frac{21}{100} \right]^2$$

$$A = 8000 \left[\frac{20 + 21}{20} \right]^2$$

$$A = 8000 \left[\frac{22.1}{20} \right]^2$$

$$A = \frac{8000}{1} \times \frac{22.1}{20} \times \frac{22.1}{20}$$

$$A = 442 \times 22.1$$

$$A = 9768.2$$

$$\begin{aligned} \text{C.I.} &= \text{Amount} - \text{Principal} \\ &= 9768.2 - 8000 \end{aligned}$$

$$\text{C.I.} = 1768.2$$

1768.2 rupees compound interest will she have to pay.

Practice Set 14.2

1. On the construction work of a flyover bridge there were 320 workers initially. The number of workers were increased by 25% every year. Find the number of workers after 2 years.

⇒ Given:

Now,

$$P = 320, R = 25\%, N = 2 \text{ years}$$

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$A = 320 \left[1 + \frac{25}{100} \right]^2 \quad 156.25$$

$$A = 320 \left[1 + \frac{1}{4} \right]^2$$

$$A = 320 \left[\frac{4+1}{4} \right]^2$$

$$A = 320 \times \left[\frac{5}{4} \right]^2$$

$$A = \overset{80}{320} \times 5 \times 5$$

$$A = 100 \times 5$$

$$A = 500 \text{ workers,}$$

The number of workers after 2 years is 500

2. A shepherd has 200 sheep with him. Find the number of sheep with him after the number of ~~workers~~ 3 years if the increase in number of sheep is 8% every year.

⇒ Given :

Here,

$P = 200$ sheep, $R = 8\%$, $N = 3$ years

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$A = 200 \left[1 + \frac{8}{100} \right]^3$$

$$A = 200 \left[\frac{25+2}{25} \right]^3$$

$$A = 200 \left[\frac{27}{25} \right]^3$$

$$A = \frac{200 \times 27 \times 27 \times 27}{25 \times 25 \times 25} = \frac{216 \times 27 \times 27}{25 \times 25}$$

$$A = \frac{5832 \times 27}{25 \times 25}$$

$$A = \frac{157464}{625}$$

$$A = 251.94$$

3. In a forest there are 40000 trees. Find the expected number of trees after 3 years if the objective is to increase the number at the rate 5% per year.

⇒ Given:

Here,

$P = 40000$ trees, $R = 5\%$, $N = 3$ years

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$A = 40000 \left[1 + \frac{5}{100} \right]^3$$

$$A = 40000 \left[\frac{1}{1} + \frac{1}{20} \right]^3$$

$$A = 40000 \left[\frac{20 + 1}{20} \right]^3$$

$$A = 40000 \left[\frac{21}{20} \right]^3$$

$$A = \frac{40000}{20} \times 21 \times 21 \times 21 = 5 \times 21 \times 21 \times 21$$

$$A = 105 \times 21 \times 21$$

$$= 2205 \times 21$$

$$A = 46305 \text{ trees}$$

4. The cost price of a machine is 250000. If the rate of depreciation is 10% per year find the depreciation in price of the machine after two years.

⇒ $P = 2,50,000$, $R = -10\%$ (because rate is depreciation) and $N = 2$ years

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$A = 250000 \left[1 + \frac{(-10)}{100} \right]^2$$

$$A = 250000 \left[\frac{-100 + 100}{100} \right]^2$$

$$A = 250000 \left[\frac{90}{100} \right]^2$$

$$A = \frac{250000 \times 9 \times 9}{10 \times 10} = 2500 \times 9 \times 9$$

$$A = 22500 \times 9$$

$$A = 202500$$

$$\text{Depreciation price} = \text{Principle} - \text{Amount} \\ = 250000 - 202500$$

$$\text{Depreciation price} = 47500 \text{ rupees.}$$

5. Find the compound interest if the amount of a certain principal after two years is ₹4036.80 at the rate of 16 p.c.p.a.

⇒ Given: $A = 4036.80$, $r = 16 \text{ p.c.p.a.}$
 $N = 2 \text{ years}$, $P = ?$

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$4036.80 = P \left[1 + \frac{16}{100} \right]^2$$

$$4036.80 = P \left[\frac{4+25}{25} \right]^2$$

$$4036.80 = P \left[\frac{29}{25} \right]^2$$

$$4036.80 = P \times \frac{29}{25} \times \frac{29}{25}$$

$$4036.80 = P \times \frac{841}{625}$$

$$P = \frac{4036.80 \times 625}{841}$$

$$P = 4.8 \times 1025$$

$$P = 3000$$

Compound interest = $A - P$

$$= 4036.80 - 3000$$

$$\text{Compound interest} = 1036.80 \text{ rupees.}$$

6. A loan of ₹ 15000 was taken on compound interest. If the rate of compound interest is 12 p.c.p.a. find the amount to settle the loan after 3 years.

⇒ Here.

$$P = 15000, R = 12\%, N = 3 \text{ years}$$

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$A = 15000 \left[1 + \frac{12}{100} \right]^3$$

$$A = 15000 \left[\frac{25+3}{25} \right]^3$$

$$A = 15000 \left[\frac{28}{25} \right]^3$$

$$A = \frac{15000 \times 28 \times 28 \times 28}{25 \times 25 \times 25} = \frac{24 \times 28 \times 28 \times 28}{25}$$

$$A = \frac{672 \times 28 \times 28}{25}$$

$$A = \frac{18816 \times 28}{25}$$

$$A = \frac{526848}{25}$$

$$A = 21073.92$$

$$A = 21073.92$$

7. A principal amounts to ₹ 13924 in 2 years by compound interest at 18 p.c.p.a. Find the principal.

⇒ Given:

$P = ?$, $A = 13924$ rupees, $N = 2$ years
 $R = 18\%$.

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$13924 = P \left[1 + \frac{18}{100} \right]^2$$

$$13924 = P \left[\frac{50 + 9}{50} \right]^2$$

$$13924 = P \left[\frac{59}{50} \right]^2$$

$$13924 = P \left[\frac{59 \times 59}{50 \times 50} \right]$$

$$13924 = P \times \frac{3481}{2500}$$

$$P = \frac{13924 \times 2500}{3481}$$

$$P = 4 \times 2500$$

$$P = 10000 \text{ rupees.}$$

8 The population of a suburb is 16000. Find the rate of increase in the population if the population after two years is 17640.

⇒ Given :

$$P = 16000, A = 17640, R = ?, N = 2 \text{ years}$$

$$A = P \cdot \left[1 + \frac{R}{100} \right]^N$$

$$17640 = 16000 \left[1 + \frac{R}{100} \right]^2$$

$$17640 = 16000 \left[\frac{100 + R}{100} \right]^2$$

$$17640 = 16000 \times \frac{100 + R}{100}^2$$

$$17640 = 16000 + \frac{R^2}{100}$$

$$17640 = 16000 + R^2$$

$$17640 - 16000 = R^2$$

$$1640 = R^2$$

$$\frac{17640}{16000} = \left(1 + \frac{R}{100} \right)^2$$

$$\therefore \sqrt{\left(\frac{42}{40} \right)^2} = \sqrt{\left(1 + \frac{R}{100} \right)^2}$$

$$\therefore \frac{42}{40} = 1 + \frac{R}{100}$$

$$\therefore \frac{42}{40} - \frac{1}{1} = \frac{R}{100}$$

$$\frac{42-40}{42} = \frac{R}{100}$$

$$\therefore \frac{2}{40} = \frac{R}{100}$$

$$\frac{2 \times 100}{40} = R$$

$$\boxed{R = 5}$$

$\therefore$ The population of subud increases 5% every year.

Q In how many years ₹ 700 will amount to ₹ 847 at a compound interest rate of 10 p.c.p.a.

2) Given:

$$P = 700 \text{ rupees} \quad R = 10 \text{ p.c.p.a.} \quad A = 847 \text{ ₹}$$

$N = ?$

$$A = P \left[1 + \frac{R}{100} \right]^N$$

$$847 = 700 \left[1 + \frac{10}{100} \right]^N$$

$$847 = 700 \left[\frac{10 + 1}{10} \right]^N$$

$$\frac{847}{700} = \left(\frac{11}{10} \right)^N$$

or

$$\frac{121}{100} = \left(\frac{11}{10} \right)^N$$

$$\left(\frac{11}{10} \right)^2 = \left(\frac{11}{10} \right)^N$$

$$\therefore N = 2$$

The required duration is 2 years.

10. Find the difference between simple interest and compound interest on ₹ 20000 at 8 p.c.p.a.

⇒ Here,

$$P = 20000 \text{ ₹}$$

$$R = 8 \text{ p.c.p.a.}$$

$$N = 2 \text{ years.}$$

$$\text{Simple interest} = \frac{P \times N \times R}{100}$$

$$= \frac{20000 \times 2 \times 8}{100}$$

$$= 400 \times 8$$

$$\text{Simple interest} = 3200 \text{ ₹.}$$

$$A = P \times \left(1 + \frac{R}{100}\right)^N$$

$$A = 20000 \times \left(1 + \frac{8}{100}\right)^2$$

$$A = 20000 \times \left(\frac{108}{100}\right)^2$$

$$A = \frac{20000 \times 108 \times 108}{100 \times 100}$$

$$A = 2 \times 108 \times 108$$

$$A = 23328 \text{ ₹.}$$

$$C.I. = A - P$$

$$= 23328 - 20000$$

$$C.I. = 3328 \text{ ₹.}$$

Difference between compound interest and simple interest

$$\therefore 3328 - 3200 = 128 \text{ rs.}$$